

OBJECTIVE	PhD candidate in Mathematics, specializing in financial stochastic processes and quantitative modeling. My research develops pricing tools for Catastrophe bonds, financial instruments that help economies absorb disaster losses by sharing risk across international capital markets.			
EDUCATION	Ph.D. in Mathematics	2022 - Present		
	<i>North Dakota State University</i>	Fargo, ND, USA		
	• Advisor: Prof. Dr. Indranil SenGupta, Co-Advisor: Prof. Nikita Barabanov • Dissertation: “Catastrophe Bonds Valuation under Jump-Diffusion Models with Domestic and Foreign Interest Rates: A Multi-Currency Analysis” • Relevant Coursework: Stochastic Processes, Measure Theory, Machine Learning, Neural Networks (Mathematical Theory). • Current CGPA: 4/4			
	M.Sc. (Research) in Mathematics	2018 - 2020		
	<i>Manav Rachna University</i>	Haryana, India		
	• Master’s Thesis: “Wavelet Transform and its Applications” • Advisor: Dr. Aparna Vyas • First Class			
	B.Sc. (Hons.) in Mathematics	2014 - 2017		
	<i>University of Delhi</i>	New Delhi, India		
PUBLICATIONS	• First Class			
	1. (Submitted) Ridhima Nehra and I. SenGupta (2026), “The valuation of catastrophe bonds under domestic and foreign interest rates with jumps”.			
	2. (Submitted) Ridhima Nehra and I. SenGupta (2026), “The Catastrophe Bond Price under the Jump Diffusion Model: A Multi-Currency Comparison”.			
	3. Ridhima Nehra , Sushma Dave, Pratik Jagtap. “Mathematical Modeling and Surface Response Curves for Green Synthesized Nanomaterials and Their Application in Dye Degradation.” <i>Book Chapter in Photocatalytic Degradation of Dyes</i> , Elsevier, 2021.			
	4. Ridhima Nehra , Rojin Joseph, and Aparna Vyas. “Comparison of Image Denoising Using Wavelet Transform and Wavelet Packet Transform.” <i>International Conference Proceedings</i> , Tokyo Summit II, 2020.			
	• Graduate Student Research Award: North Dakota State University, 2026. • Travel Award: American Mathematical Society and Department of Mathematics, NDSU, for the Spring Southeastern Sectional Meeting, March 2026. • Excellence in Research: Manav Rachna University, 2020. • Best Paper Award: “Comparison of Image Denoising Using Wavelet Transform and Wavelet Packet Transform”, <i>Tokyo Summit II: 2nd International Conference on Innovative Studies of Contemporary Sciences</i>, 2020. • Best Paper Presentation Award: “A Review on Applications of Wavelets in Chemistry”, Oral Presentation Category, <i>Green and Sustainable Chemistry Conference (GSCC) 2019</i>.			
	AWARDS AND FUNDING			
RESEARCH INTERESTS	Financial Mathematics, Stochastic Processes, Jump-Diffusion Models, Mathematical Modeling, Machine Learning, Large Language Model Fine-Tuning, Time Series Forecasting, Monte Carlo Simulation, Differential Equations, Wavelets			
PRESENTATIONS	• “American Mathematical Society 2026 Spring Southeastern Sectional Meeting” at Georgia Southern University-Armstrong Campus, Savannah, GA, March 28-29, 2026.			
	• “2nd International Conference on Innovative Studies of Contemporary Sciences”, Tokyo, Japan, August 17-19, 2020.			
	• “Green and Sustainable Chemistry Conference GSCC”, Manav Rachna University, India, Nov 7-8, 2019.			
	• “Analysis, Applied Math & Geometry” Department Seminar, NDSU, March 24, 2026.			
	• “Graduate Research Days”, NDSU, April 09, 2026.			

RESEARCH
PROJECTS

Catastrophe Bond Price using Jump-Diffusion Model:

- Extended the diffusion model of Catastrophe bond price by incorporating a jump-diffusion model into exchange rate and interest rate processes, and derived closed-form pricing formulas for zero-coupon bonds, FX call options and Catastrophe bond price.
- Validated the model using Monte Carlo simulation, conducted goodness-of-fit tests and performed parameter sensitivity analysis across jump intensity, jump size, volatility, and mean-reversion speed.

Multi-Currency Extension of the Jump-Diffusion Model:

- Extended the jump-diffusion Cat Bond price to four currency pairs (USD/EUR, USD/INR, USD/BRL, USD/JPY).
- Conducted Monte Carlo simulation, parameter calibration and sensitivity analysis across differing volatility and economies.

Mental Health Emotion Detection Using Open-Source LLMs:

- Evaluated four transformer-based pipelines (fine-tuned BERT, fine-tuned DistilBERT, and hybrid embeddings with ML classifiers) for multi-label emotion classification on the GoEmotions dataset.
- Applied threshold tuning, class balancing, and classifier stacking to address class imbalance; conducted hyperparameter optimization and benchmarked models on F1, precision.

INTERNSHIPS

Summer Research Intern, Indian Institute of Science Education and Research (IISER) | Kolkata, India Summer 2019

- Supervised by Dr. Ratikanta Behera, Assistant Professor, IISER Kolkata
- Applied wavelet transforms and mathematical modeling for data analysis, developing algorithms for image processing and noise reduction.
- Analyzed and processed data using wavelet-based techniques to enhance accuracy and efficiency in real-world problems.

Summer Research Programme Intern, CSIR-NEIST | Online Summer 2020

- Supervised by Dr. Sushma Dave, Associate Professor, JIET Group of Institutions, Rajasthan
- Developed mathematical models for nanomaterial synthesis and photocatalytic dye degradation.
- Used statistical methods and Design of Experiments (DoE) to create surface response curves and optimize dye degradation parameters.

TEACHING
EXPERI-
ENCE**Primary Instructor, Department of Mathematics, NDSU**

- Spring & Fall (2025, 2026), Summer 2026: Basic Linear Algebra - MATH 129.

Teaching Assistant, Department of Mathematics, NDSU

- Fall 2024: TA for Calculus I (MATH 165) and Calculus II (MATH 166) - quizzes, exams, grading. Tutor at ACE across nine courses (MATH 98, 103-105, 128,129, 165,166, 259, 265,266).
- Spring 2024: TA for Calculus I (MATH 165): quizzes, exams, grading. Grader: MATH 329 (Intermediate Linear Algebra).
- Fall 2023: TA for Calculus I (MATH 165): quizzes, exams, grading. Grader: MATH 266 (Intro to Differential Equations).
- Spring 2023: TA at Emporium (MATH 103, 105, 107): quizzes, exams, grading. Grader: MATH 266.
- Fall 2022: TA at Emporium (MATH 103, 105, 107): quizzes, exams, grading. Tutor at ACE across nine courses (MATH 98, 103-105, 128,129, 165,166, 259, 265,266).

CERTIFICATIONS

- Course on **CIMER: Entering Mentoring**, an evidence-based research mentor training program focused on developing effective mentoring practices, Sep-Dec 2025.
- Course on “**Python for Financial Analysis and Algorithmic Trading**”, by Jose Portilla, Udemy (2024).
- **Post-Graduate Diploma in Applied Statistics (PGDAST)** from IGNOU, New Delhi, India 2021-2022 [Training in *Data Analysis, Statistical Inference & Techniques, Descriptive & Industrial Statistics*]
- Course on “**Geo-informatics for Biodiversity Conservation Planning**” conducted by IIRS, Indian Space Research Organization (ISRO) in 2021.
- Course on “**Quantum Fields, Geometry and Representation Theory**”, International Centre for Theoretical Sciences, *Tata Institute of Fundamental Research*, July 2021 .

CONFERENCES AND WORK- SHOPS AT- TENDED	• Attended “Great Plains Combinatorics Conference 2026” organized by the Department of Mathematics, University of Kansas, Lawrence, from May 2-3, 2026. • Attended “AMS Spring Central Sectional Meeting” at North Dakota State University, Fargo, from April 18-19, 2026. • Regularly attend departmental Colloquium and seminars at NDSU, including Algebra & Discrete Mathematics, Analysis & Geometry, and Applied Mathematics (2022-present). • Attended International Workshop on “Wavelets and its Applications: Image Processing, Data Science and PDEs (WAIDP-2021)”, organized by the Department of Mathematics, IIT Indore, from December 06th -10th, 2021. • Attended International Webinar on “Wavelet Approximation in Orlicz Spaces” organized by the Department of Mathematics, Manav Rachna University, Faridabad, on September 27, 2021. • Attended an invited talk on ”Optimization Algorithms” by Dr. Aparna Mehra, Professor from IIT Delhi, organized by the Department of Mathematics, Manav Rachna University on 25 September, 2019. • Attended an invited talk on ”Scope for Mathematicians in the field of Actuarial Sciences” organized by the Department of Mathematics, Manav Rachna University on 1 March, 2019. • Attended an invited talk on ”Role of Mathematical Techniques in Medical Decision Making Processes” organized by the Department of Mathematics, MRIIRS, Faridabad on 25th January 2019. • Attended “International Conference On Applicable Analysis” organized by the Department of Mathematics, Shaheed Bhagat Singh College, dated 08-11 February 2017.
EXTRACURRICULAR ACTIVITIES	• Math Genealogy Project: Contributed to collecting and tracking the academic genealogy of mathematicians, adding new entries and updating existing records in the database (2024-2026). • Math-In : Assisted undergraduate students with assignment review and concept clarification in preparation for final examinations (2023-2026). • Served as Cultural Secretary of the Department of Mathematics, Shaheed Bhagat Singh College (2016-17). • Won First Position in Shot Put at Shaheed Bhagat Singh College (2016, 2017). • Organized multiple events on behalf of the Department of Mathematics, Manav Rachna University (2019, 2020). • Coordinated and organized various cultural and departmental festivals at Shaheed Bhagat Singh College (2015-2017).
MEMBERSHIP	• American Mathematical Society (AMS), 2022-present • Society for Industrial and Applied Mathematics (SIAM), 2024-present
TECHNICAL SKILLS	Python, Mathematica, MATLAB, R, SPSS, Excel, $\text{\LaTeX}$, Scikit-learn, Pandas, NumPy, QuantPy, PyTorch, Hugging Face Transformers
REFERENCES	Available upon request.